TEST CASE DOCUMENT TABLES (Aleksandra, Elias, Shivang, Izzy)

Find the Thimble

Use Case

Primary Actor: User (Player), Secondary Actor: Computer

1. Description The user plays Find the Thimble against the computer. The computer hides a thimble in one of two hands, and the user guesses which hand holds it. Players first agree on a best out of value. The first to reach the win threshold wins the game.

2. Preconditions

- The Game of Games main menu has been displayed.
- The user has selected Find the Thimble from the menu.

3. Main Success Scenario (Flow of Events)

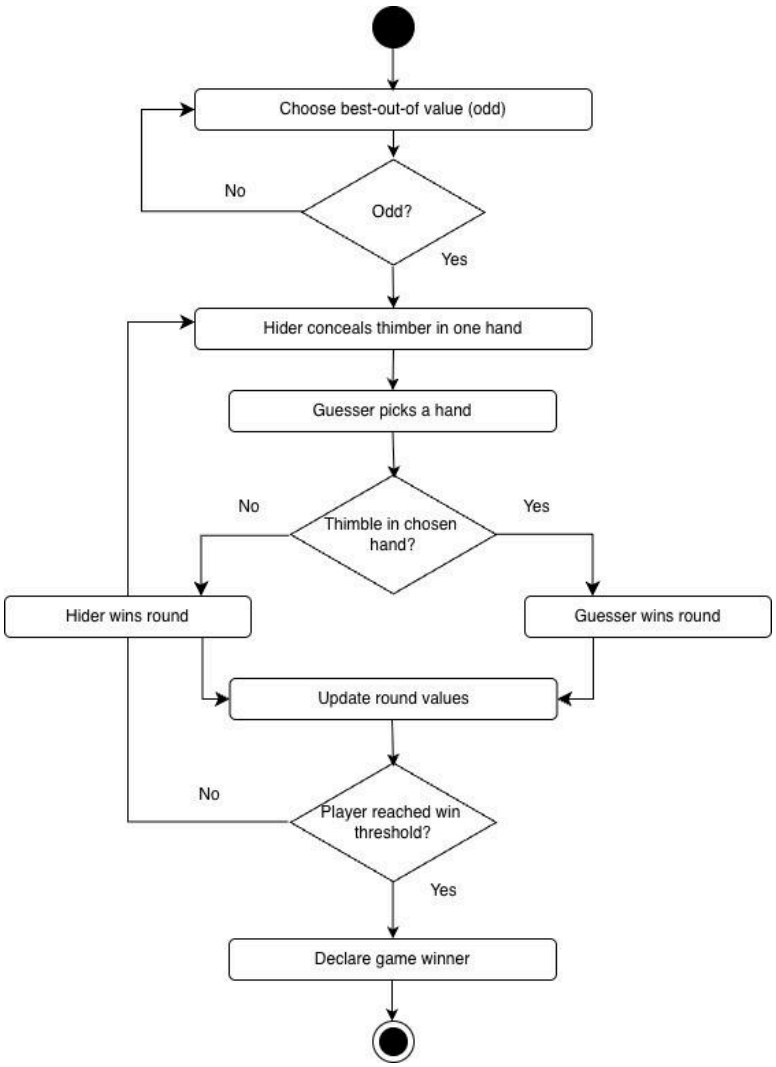
1. **Enter Best-Out-Of Value:** The computer prompts: "Enter the best-out-of value (must be an odd number):" The user enters a valid odd number (e.g., 5). The win threshold is set to (value + 1) / 2. For 5, the threshold is 3.
2. **Hide the Thimble:** The computer hides the thimble in one hand.
3. **User Guesses:** The computer displays: "Guess which hand holds the thimble. Enter L for left or R for right:" The user enters L or R.
4. **Reveal Result:** The computer reveals which hand held the thimble. If the guess is correct: "Correct! You win this round." If incorrect: "Wrong! The computer wins this round."
5. **Update Scores:** The computer updates and displays the round scoreboard, e.g., "You: 1, Computer: 0".
6. **Continue or End:** If neither player has reached the win threshold, return to Step 2. Otherwise, proceed to Step 7.
7. **Declare Game Winner:** The computer announces: "You win Find the Thimble!" or "The computer wins Find the Thimble!" The game scoreboard is updated and displayed, then control returns to the Game of Games main menu.

4. Errors

- Invalid Best-Out-Of Value:
    - If the user enters an even number or non-numeric input, the computer displays: "Invalid entry. The value must be an odd whole number. Please try again:" and returns to Step 1.
- Invalid Hand Guess:

- If the user enters anything other than L or R, the computer displays: "Invalid input. Please enter L for left or R for right:" and re-prompts Step 3.

Activity Diagram



Errors

Sunny Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Valid odd best-out-of value accepted and win threshold set correctly. Input: 5	Win threshold set to 3, round 1 begins.			
	2	The user wins the round by correctly guessing the right hand. Input: R (when thimble is in right hand)	"Correct! You win this round." Score displayed as You: 1, Computer: 0.			
	3	The user wins the round by correctly guessing the left hand. Input: L (when thimble is in left hand)	"Correct! You win this round." Score updated accordingly.			
	4	Computer wins round when the user guesses incorrectly. Input: L (when thimble is in right hand)	"Wrong! The computer wins this round." Score displayed as You: 0, Computer: 1.			
	5	The user wins the game upon reaching the win threshold. Win 3	"You win, Find the Thimble!" Game			

		rounds in a best-of-5.	scoreboard updated.			
	6	The computer wins the game upon reaching the win threshold. Allow the computer to win 3 rounds in a best-of-5.	"The computer wins, Find the Thimble!" Game scoreboard updated.			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Valid odd best-out-of value accepted and win threshold set correctly. Input: 5	Win threshold set to 3, round 1 begins.	Win threshold set to 3, round 1 begins.	X	
X	2	The user wins the round by correctly guessing the right hand. Input: R (when thimble is in right hand)	"Correct! You win this round." Score displayed as You: 1, Computer: 0.	"Correct! You win this round." Score displayed as You: 1, Computer: 0.	X	
X	3	The user wins the round by correctly guessing the left hand. Input: L (when thimble is in left hand)	"Correct! You win this round." Score updated accordingly.	"Correct! You win this round." Score updated accordingly.	X	
X	4	The computer wins round when the user guesses incorrectly. Input: L	"Wrong! The computer wins this round."	"Wrong! The computer wins this round." Score	X	

		(when thimble is in right hand)	Score displayed as You: 0, Computer: 1.	displayed as You: 0, Computer: 1.		
X	5	The user wins the game upon reaching the win threshold. Win 3 rounds in a best-of-5.	"You win, Find the Thimble!" Game scoreboard updated.	"You win, Find the Thimble!" Game scoreboard updated.	X	
X	6	The computer wins the game upon reaching the win threshold. Allow the computer to win 3 rounds in a best-of-5.	"The computer wins, Find the Thimble!" Game scoreboard updated.	"The computer wins, Find the Thimble!" Game scoreboard updated.	X	

Rainy Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Even best-out-of value rejected. Input: 4	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.			

	2	Non-numeric best-out-of value rejected. Input: abc	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.			
	3	Invalid hand guess character rejected. Input: X	"Invalid input. Please enter L for left or R for right:" Re-prompted.			
	4	Lowercase hand guess rejected. Input: l	"Invalid input. Please enter L for left or R for right:" Re-prompted.			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Even best-out-of value rejected. Input: 4	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	X	
X	2	Non-numeric best-out-of value rejected. Input: abc	"Invalid entry. The value must be an odd whole number.	"Invalid entry. The value must be an odd whole number. Please try	X	

			Please try again:" Re-prompted.	again:" Re-prompted.		
X	3	Invalid hand guess character rejected. Input: X	"Invalid input. Please enter L for left or R for right:" Re-prompted.	"Invalid input. Please enter L for left or R for right:" Re-prompted.	X	
X	4	Lowercase hand guess rejected. Input: l	"Invalid input. Please enter L for left or R for right:" Re-prompted.	"Invalid input. Please enter L for left or R for right:" Re-prompted.	X	

Coin Flip

Use Case

Primary Actor: User (Caller), Secondary Actor: Computer

1. Description The user plays Coin Flip against the computer. Each round, the user calls heads or tails, and the computer flips a coin. Players first agree on a best out of value. The first player to reach the win threshold wins the game.

2. Preconditions

- The Game of Games main menu has been displayed.
- The user has selected Coin Flip from the menu.

3. Main Success Scenario (Flow of Events)

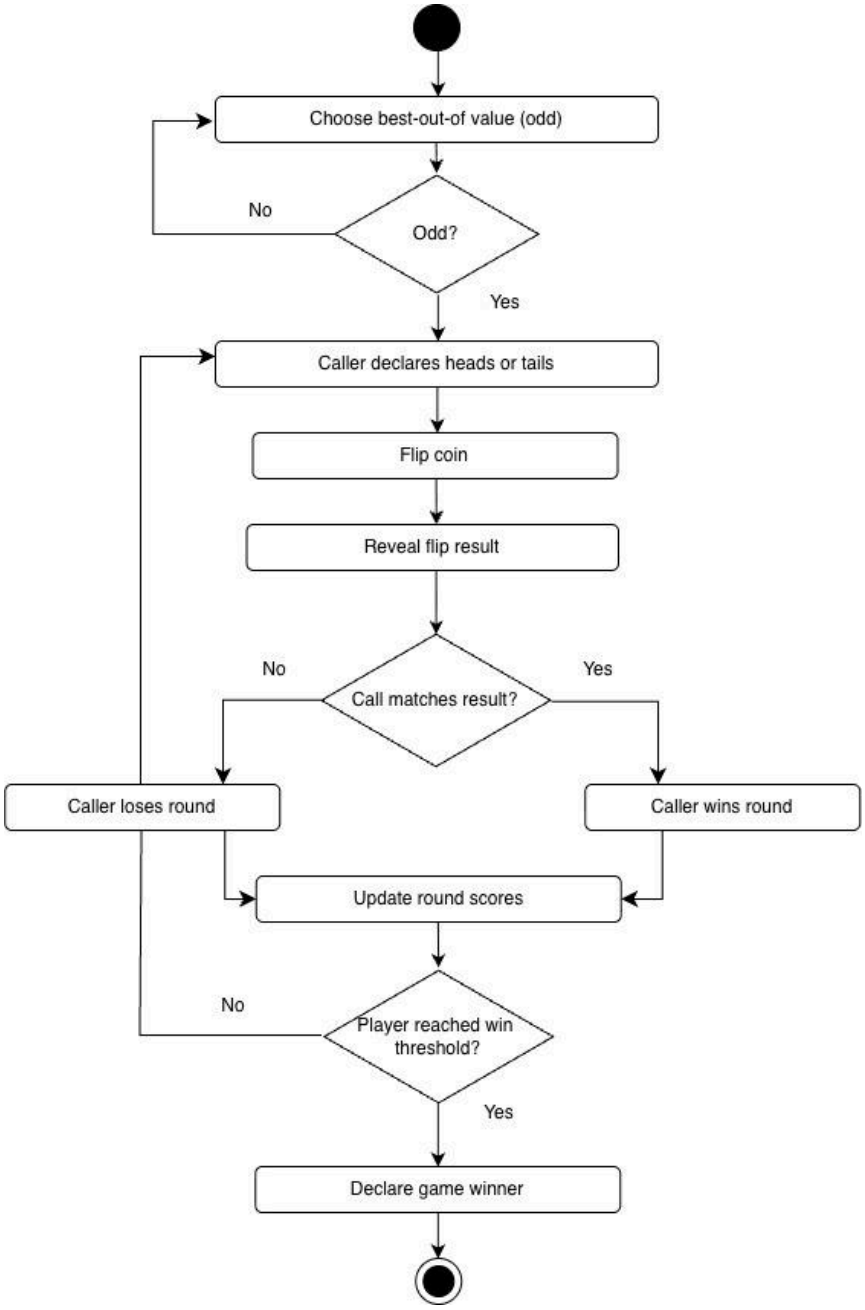
1. **Enter Best-Out-Of Value:** The computer prompts: "Enter the best-out-of value (must be an odd number):" The user enters a valid odd number (e.g., 7). The win threshold is set to (value + 1) / 2. For 7, the threshold is 4.
2. **User Calls the Flip:** The computer prompts: "Call it, enter H for heads or T for tails:" The user enters H or T.
3. **Flip the Coin:** The computer flips the coin and displays the result, e.g., "The coin landed on: Heads."
4. **Reveal Round Result:** If the call matches the result: "Correct call! You win this round." If the call does not match: "Wrong call! The computer wins this round."

5. **Update Scores:** The computer updates and displays the round scoreboard, e.g., "You: 2, Computer: 1".
6. **Continue or End:** If neither player has reached the win threshold, return to Step 2. Otherwise, proceed to Step 7.
7. **Declare Game Winner:** The computer announces: "You win Coin Flip!" or "The computer wins Coin Flip!" The game scoreboard is updated and displayed, then control returns to the Game of Games main menu.

4. Errors

- Invalid Best-Out-Of Value:
  - If the user enters an even number or non-numeric input, the computer displays: "Invalid entry. The value must be an odd whole number. Please try again:" and returns to Step 1.
- Invalid Coin Call:
  - If the user enters anything other than H or T, the computer displays: "Invalid input. Please enter H for heads or T for tails:" and re-prompts Step 2.

Activity Diagram



Errors

Sunny Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Valid odd best-out-of value accepted and win threshold set correctly. Input: 7	Win threshold set to 4,round 1 begins, user prompted for H or T.	Win threshold set to 4, round 1 begins, user prompted for H or T.	X	
	2	The user wins the round with the correct heads call. Input: H (when coin lands Heads)	"Correct call! You win this round." Score updated.	"Correct call! You win this round." Score updated.	X	
	3	The user wins the round with the correct tails call. Input: T (when coin lands Tails)	"Correct call! You win this round." Score updated.	"Correct call! You win this round." Score updated.	X	
	4	The computer wins round when the user call is incorrect. Input: H (when coin lands Tails)	"Wrong call! The computer wins this round." Score updated.	"Wrong call! The computer wins this round." Score updated.	X	
	5	The coin result is displayed before the round winner is announced. Input: H	"The coin landed on: Heads." shown before result announcemen t.	"The coin landed on: Heads." shown before result announcement.	X	

	6	The user wins the game upon reaching the threshold. Win 4 rounds in a best-of-7.	"You win, Coin Flip!" Scoreboard updated.	"You win, Coin Flip!" Scoreboard updated.	X	

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Valid odd best-out-of value accepted and win threshold set correctly. Input: 7	Win threshold set to 4,round 1 begins, user prompted for H or T.	Win threshold set to 4, round 1 begins, user prompted for H or T.	X	
X	2	The user wins the round with the correct heads call. Input: H (when coin lands Heads)	"Correct call! You win this round." Score updated.	"Correct call! You win this round." Score updated.	X	
X	3	The user wins the round with the correct tails call. Input: T (when coin lands Tails)	"Correct call! You win this round." Score updated.	"Correct call! You win this round." Score updated.	X	
X	4	The computer wins round when the user call is incorrect. Input: H (when coin lands Tails)	"Wrong call! The computer wins this round." Score updated.	"Wrong call! The computer wins this round." Score updated.	X	
X	5	The coin result is displayed before the round winner	"The coin landed on: Heads." shown before	"The coin landed on: Heads." shown	X	

		is announced. Input: H	result announcement.	before result announcement.		
X	6	The user wins the game upon reaching the threshold. Win 4 rounds in a best-of-7.	"You win, Coin Flip!" Scoreboard updated.	"You win, Coin Flip!" Scoreboard updated.	X	

Rainy Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Even best-out-of value rejected. Input: 6	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.			
	2	Non-integer best-out-of value rejected. Input: 3.5	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.			
	3	Invalid coin call character rejected. Input: W	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.			

	4	Lowercase coin call rejected. Input: h	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Even best-out-of value rejected. Input: 6	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	X	
X	2	Non-integer best-out-of value rejected. Input: 3.5	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	"Invalid entry. The value must be an odd whole number. Please try again:" Re-prompted.	X	
X	3	Invalid coin call character rejected. Input: W	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.	X	
X	4	Lowercase coin call rejected. Input: h	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.	"Invalid input. Please enter H for heads or T for tails:" Re-prompted.	X	

Guess the Number

Sunny Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Guess the Number, valid range and guess count accepted when guesses are at most half the range. Input: Range: 10, Guesses: 5	The computer randomly generates a number to guess based on the range. Number: 7			
	2	Guess the Number, the user makes an incorrect guess while still having guesses. Guesses left: 5	The number of guesses decreases, indicated by the following message: "Wrong guess! Guesses left: 4"			
	3	Guess the Number, the user makes an incorrect guess while not having guesses left in a round Guesses left: 1	Indicates the user lost with "Wrong! The computer wins this round."			
	4	Guess the Number, user wins round with correct guess within the allowed number of guesses. Enter the correct	Indicates the user won with "Correct! You win this round" and starts the next run.			

		number within the guess				

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Guess the Number, valid range and guess count accepted when guesses are at most half the range. Input: Range: 10, Guesses: 5	The computer randomly generates a number to guess based on the range. Number: 7	The computer randomly generates a number to guess based on the range. Number: 7	X	
X	2	Guess the Number, the user makes an incorrect guess while still having guesses. Guesses left: 5	The number of guesses decreases, indicated by the following message: "Wrong guess! Guesses left: 4"	The number of guesses decreases, indicated by the following message: "Wrong guess! Guesses left: 4"	X	
X	3	Guess the Number, the user makes an incorrect guess while not having guesses left in a round Guesses left: 1	Indicates the user lost with "Wrong! The computer wins this round."	Indicates the user lost with "Wrong! The computer wins this round."	X	
X	4	Guess the Number, user wins round with correct guess within the allowed number of guesses. Enter the correct	Indicates the user won with "Correct! You win this round" and starts the next run.	Indicates the user won with "Correct! You win this round" and starts the next run.	X	

		number within the guess				
X	5	Guess the Number, the final round is won by the player	Indicates the user won with “You win Guess the Number!”.	Indicates the user won with “You win Guess the Number!”.	X	
X	6	Guess the Number, the final round is won by the computer	Indicates the computer won with “The computer wins Guess the Number!”.	Indicates the computer won with “The computer wins Guess the Number!”.	X	

Rainy Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Guess the Number, the user types in a non-integer amount for the range Guess: Ten	The computer prompts the user to try inputting a different amount: “Invalid range. Please try again:”			
	2	Guess the Number, valid range and guess count is over half of the range. Input: Range: 10, Guesses: 7	The computer prompts the user to choose a different number: “The number of guesses is over half the range. Please type a different number”			

	3	Guess the Number, the user types in a non-integer guess Guess: Five	The computer prompts the user to try another guess: “Invalid guess. Please try again”			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Guess the Number, the user types in a non-integer amount for the range Guess: Ten	The computer prompts the user to try inputting a different amount: “Invalid range. Please try again:”	The computer prompts the user to try inputting a different amount: “Invalid range. Please try again:”	X	
X	2	Guess the Number, valid range and guess count is over half of the range. Input: Range: 10, Guesses: 7	The computer prompts the user to choose a different number: “The number of guesses is over half the range. Please type a different number”	The computer prompts the user to choose a different number: “The number of guesses is over half the range. Please type a different number”	X	
X	3	Guess the Number, the user types in a non-integer guess Guess: Five	The computer prompts the user to try another guess: “Invalid guess. Please try again”	The computer prompts the user to try another guess: “Invalid guess. Please try again”	X	

Even and Odd

Sunny Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORREC T	INCORREC T
	1	Even and Odd, player selects Even role; computer is assigned Odd. Input: **E**	Roles assigned for the entire game; best-out-of prompt displayed: “You are even.”			
	2	Even and Odd, valid odd best-out-of value accepted and win threshold set correctly. Input: **7**	Win threshold set to 4; round 1 begins.			
	3	Even and Odd, even player wins round when sum of both numbers is even. Enter an integer which, based on the computer's pick, results in an even sum.	Even player's score incremented; result shown e.g. "2+4=6 is even, you win this round."			
	4	Even and Odd, odd player wins round when sum of both numbers is odd. Enter an integer which, based on the	Odd player's score incremented; result shown e.g. "3+2=5 is odd, computer wins this round."			

		computer's pick, results in an odd sum.				
	5	Even and Odd, both players' numbers and their sum are displayed each round. Input: **3**	Both numbers and computed sum shown with the round winner announced.			
	6	Even and Odd, user wins the game upon reaching the threshold. Win enough rounds to meet the threshold.	"You win Even and Odd!" Scoreboard updated; returns to main menu.			
	7	Even and Odd, computer wins the game upon reaching threshold. Allow computer to win enough rounds.	"The computer wins Even and Odd!" Scoreboard updated; returns to main menu.			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORREC T	INCORREC T
X	1	Even and Odd, player selects Even role; computer is assigned Odd. Input: **E**	Roles assigned for the entire game; best-out-of prompt displayed: “You are even.”	Roles assigned for the entire game; best-out-of prompt displayed:	X	

				“You are even.”		
X	2	Even and Odd, valid odd best-out-of value accepted and win threshold set correctly. Input: **7**	Win threshold set to 4; round 1 begins.	Win threshold set to 4; round 1 begins.	X	
X	3	Even and Odd, even player wins round when sum of both numbers is even. Enter an integer which, based on the computer's pick, results in an even sum.	Even player's score incremented; result shown e.g. "2+4=6 is even, you win this round."	Even player's score incremented; result shown e.g. "2+4=6 is even, you win this round."	X	
X	4	Even and Odd, odd player wins round when sum of both numbers is odd. Enter an integer which, based on the computer's pick, results in an odd sum.	Odd player's score incremented; result shown e.g. "3+2=5 is odd, computer wins this round."	Odd player's score incremented; result shown e.g. "3+2=5 is odd, computer wins this round."	X	
X	5	Even and Odd, both players' numbers and their sum are displayed each round. Input: **3**	Both numbers and computed sum shown with the round winner announced.	Both numbers and computed sum shown with the round winner announced.	X	

X	6	Even and Odd, user wins the game upon reaching the threshold. Win enough rounds to meet the threshold.	"You win Even and Odd!" Scoreboard updated; returns to main menu.	"You win Even and Odd!" Scoreboard updated; returns to main menu.	X	
X	7	Even and Odd, computer wins the game upon reaching threshold. Allow computer to win enough rounds.	"The computer wins Even and Odd!" Scoreboard updated; returns to main menu.	"The computer wins Even and Odd!" Scoreboard updated; returns to main menu.	X	

Rainy Day Paths

Before Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORREC T	INCORREC T
	1	Even and Odd, player selects an invalid role: Role: Evenodd	Reprompt the user to try entering the role again: “Invalid role. Please try again: ”			
	2	Even and Odd, best-out-of value accepted and win threshold are invalid. Best: 20 Threshold: -1	Reprompt the user to try entering the values again: “Invalid value. Please try again: ”			

	3	Even and Odd, the player inputs an invalid integer. Number: -1	Reprompt the user to try entering the values again: “Invalid number. Please try again: ”			

After Testing

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORREC T	INCORREC T
X	1	Even and Odd, player selects an invalid role: Role: Evenodd	Reprompt the user to try entering the role again: “Invalid role. Please try again: ”	Reprompt the user to try entering the role again: “Invalid role. Please try again: ”	X	
X	2	Even and Odd, best-out-of value accepted and win threshold are invalid. Best: 20 Threshold: -1	Reprompt the user to try entering the values again: “Invalid value. Please try again: ”	Reprompt the user to try entering the values again: “Invalid value. Please try again: ”	X	
X	3	Even and Odd, the player inputs an invalid integer. Number: -1	Reprompt the user to try entering the values again: “Invalid number. Please try again: ”	Reprompt the user to try entering the values again: “Invalid number. Please try again: ”	X	

Sunny Day Paths

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	23	Guess the Number, user wins if correctly guesses Right hand. Input: R	Indicates user won with "You Won!"	"You lost!"		X
X	24	Find the Thimble, user wins if correctly guessed Left hand. Input: L	Indicates the user won with "You Won!"	"You Won!"	X	

Find the Red Thread

Use Cases for Find the Red Thread

Primary Actor: User (Player), Secondary Actor: Computer

1. Description The user plays Find the Red Thread against the computer. There are 20 spools of thread, one of which is red. Players take turns pulling a limited number of spools from the box. The player who pulls the red thread wins the game.

2. Preconditions

- The Game of Games main menu has been displayed.
- The user has selected Find the Red Thread from the menu.

3. Main Success Scenario (Flow of Events)

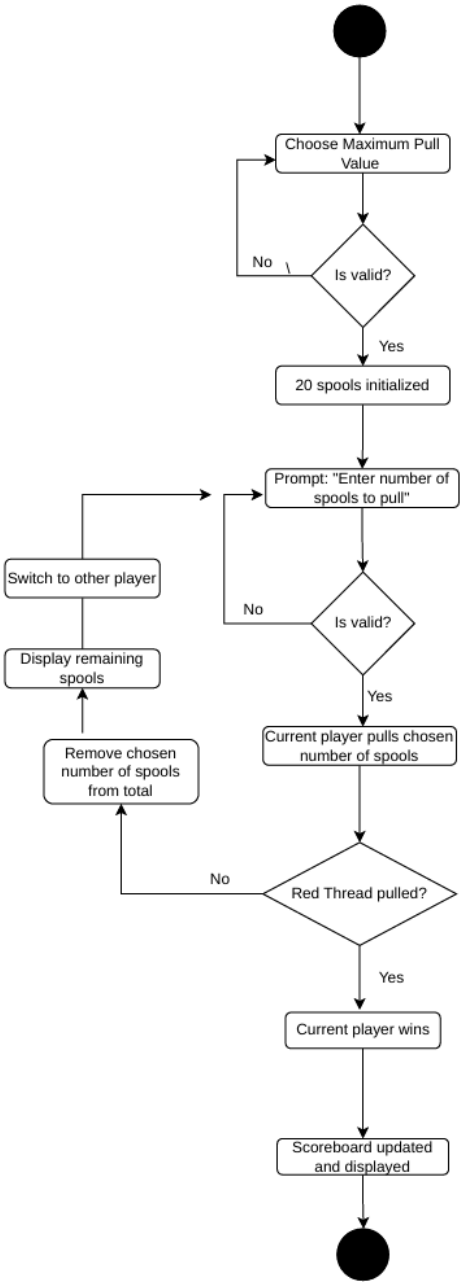
1. Choose Maximum Pull Value: The computer prompts:
   - “Enter the maximum amount of spools that can be pulled per turn (must be no more than half of 20)”
   - The user enters a valid number within the allowed range
2. **Initialize Game:**
   - The computer initializes 20 spools with one randomly designated as the red thread
3. **User Turn:**
   - The computer displays:
     "Enter the number of spools to pull (1 to [max]):”
   - The user enters a valid number within the allowed range
4. **Check for Red Thread (User):**
   - The computer randomly determines if the red thread is among the selected spools
   - If yes:
     - The computer displays: “You pulled the red thread! You win!”
     - Continue to Step 7
   - If no:
     - The pulled spools are removed from the total
     - Continue to Step 5
5. **Computer Turn:**
   - The computer selected a valid number of spools (1 to [max])
   - The computer determines if the red thread is among its selection
6. **Check for Red Thread (Computer):**
   - If yes:
     - The computer displays: “The computer pulled the red thread! The computer wins!”

        - Continue to Step 7
    - If no:
        - The pulled spools are removed from the total
        - Display the remaining spools
        - Continue to Step 3
7. **End Game:**
    - The result is displayed
    - The game scoreboard is updated and displayed
    - Control returns to the Game of Games main menu

4. Errors

- **Invalid Maximum Pull Value**
    - If the user enters a number less than 1 or greater than 10:
    "Invalid entry. The value must be between 1 and 10. Please try again:"
    - Continue to Step 1
- **Invalid Pull Amount:**
    - If the user enters a number outside the allowed range (1 to max):
    "Invalid input. Please enter a number between 1 and [max]:"
    - Re-prompt Step 3
- **Exceeding Remaining Spools:**
    - If the user attempts to pull more spools than remain:
    "Invalid move. You cannot pull more spools than remain:"
    - Re-prompt Step 3

Find The Red Thread Activity Diagram



Find The Red Thread Test Cases

Note: I had to take screenshots because it wouldn't copy over properly from Word to Google Docs

Sunny Day Paths

EXAMPLE BEFORE TESTING

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Find the Red Thread, valid max pull value accepted and game initialized with 20 spools. Input: **5**	Game starts with 20 spools; user prompted to pull 1 to 5.			
	2	Find the Red Thread, user wins when their pulled spools include the red thread. Pull a number of spools that includes the red thread's position.	"You pulled the red thread! You win!" Scoreboard updated; returns to main menu.			
	3	Find the Red Thread, computer wins when its pulled spools include the red thread. Pull spools that miss the red thread; let computer's turn find it.	"The computer pulled the red thread! The computer wins!" Scoreboard updated; returns to main menu.			
	4	Find the Red Thread, remaining spool count decreases correctly after each turn. Input: **3**	17 spools remain displayed after user pulls 3; count continues decreasing each turn.			
	5	Find the Red Thread, turns alternate correctly between user and computer. Pull spools that miss the red thread each turn.	Game alternates user → computer → user → computer until red thread is found.			

EXAMPLE AFTER TESTING

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Find the Red Thread, valid max pull value accepted and game initialized with 20 spools. Input: **5**	Game starts with 20 spools; user prompted to pull 1 to 5.	Game starts with 20 spools; user prompted to pull 1 to 5.	X	
X	2	Find the Red Thread, user wins when their pulled	"You pulled the red thread! You win!"	"You pulled the red thread! You win!"	X	

		spools include the red thread. Pull a number of spools that includes the red thread's position.	Scoreboard updated; returns to main menu.	Scoreboard updated; returns to main menu.		
X	3	Find the Red Thread, computer wins when its pulled spools include the red thread. Pull spools that miss the red thread; let computer's turn find it.	"The computer pulled the red thread! The computer wins!" Scoreboard updated; returns to main menu.	"The computer pulled the red thread! The computer wins!" Scoreboard updated; returns to main menu.	X	
X	4	Find the Red Thread, remaining spool count decreases correctly after each turn. Input: **3**	17 spools remain displayed after user pulls 3; count continues decreasing each turn.	17 spools remain displayed after user pulls 3; count continues decreasing each turn.	X	
X	5	Find the Red Thread, turns alternate correctly between user and computer. Pull spools that miss the red thread each turn.	Game alternates user → computer → user → computer until red thread is found.	Game alternates user → computer → user → computer until red thread is found.	X	

Rainy Day Paths

EXAMPLE BEFORE TESTING

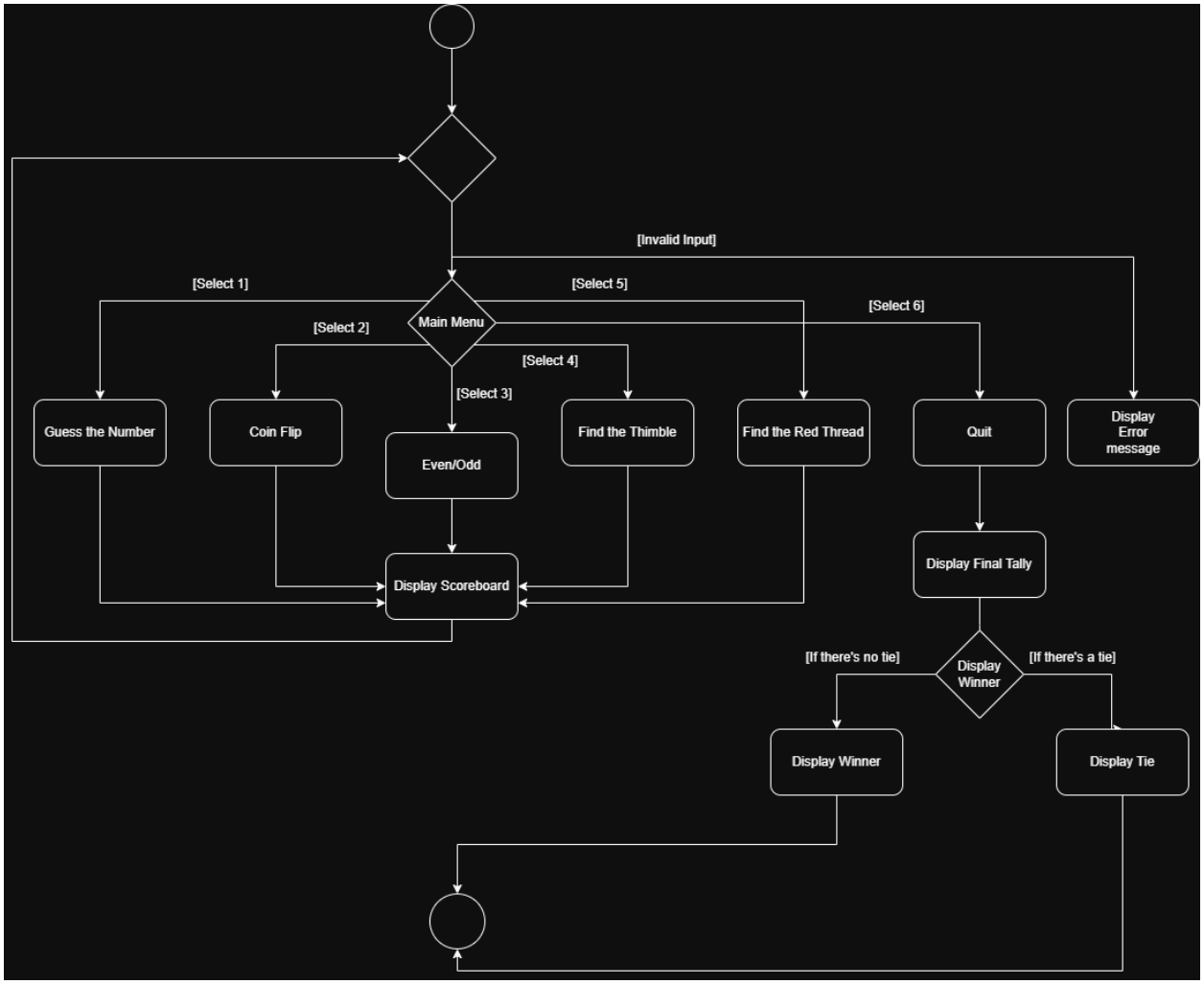
RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
	1	Find the Red Thread, max pull value greater than 10 rejected. Input: **11**	"Invalid entry. The value must be between 1 and 10. Please try again:" Re-prompted.			
	2	Find the Red Thread, max pull value of 0 rejected. Input: **0**	"Invalid entry. The value must be between 1 and 10. Please try again:" Re-prompted.			
	3	Find the Red Thread, pull amount outside the allowed range rejected. Input: **Max=5; Pull attempt: 7**	"Invalid input. Please enter a number between 1 and 5:" Re-prompted.			
	4	Find the Red Thread, pull amount exceeding remaining spools rejected.	"Invalid move. You cannot pull more spools than remain:" Re-prompted.			

		Input: **Max=10; 3 spools remain; Pull: 5**				
	5	**Find the Red Thread, non-numeric pull input rejected. Input: abc**	Error message displayed; re-prompted for a valid number.			

EXAMPLE AFTER TESTING

RUN	NUM	TEST	EXPECTED RESULT	ACTUAL RESULT	CORRECT	INCORRECT
X	1	Find the Red Thread, max pull value greater than 10 rejected. Input: **11**	"Invalid entry. The value must be between 1 and 10. Please try again:" Re-prompted.	"Invalid entry. The value must be between 1 and 10. Please try again:" Re-prompted.	X	
X	2	Find the Red Thread, max pull value of 0 rejected. Input: **0**	"Invalid entry. The value must be between 1 and [Max pull value]. Please try again:" Re-prompted.	"Invalid entry. The value must be between 1 and [Max pull value]. Please try again:" Re-prompted.	X	
X	3	Find the Red Thread, pull amount outside the allowed range rejected. Input: **Max=5; Pull attempt: 7**	"Invalid input. Please enter a number between 1 and 5:" Re-prompted.	"Invalid input. Please enter a number between 1 and 5:" Re-prompted.	X	
X	4	Find the Red Thread, pull amount exceeding remaining spools rejected. Input: **Max=10; 3 spools remain; Pull: 5**	"Invalid move. You cannot pull more spools than remain:" Re-prompted.	"Invalid move. You cannot pull more spools than remain:" Re-prompted.	X	
X	5	**Find the Red Thread, non-numeric pull input rejected. Input: abc**	Error message displayed; re-prompted for a valid number.	Error message displayed; re-prompted for a valid number.	X	

OVERALL GAME OF GAMES



Use Case

Primary Actor: User (Player) **Secondary Actor:** Computer

1. Description

The user interacts with a main menu to select and play various games against the computer. After each game, scores are updated and displayed. The session ends when the user chooses to quit, at which point a final winner is declared .

2. Preconditions

- The program has been launched and the environment is ready for text input.

3. Main Success Scenario (Flow of Events)

1. **Display Welcome:** The computer displays a welcome message and the "Game of Games" title.

2. **Display Main Menu:** The computer presents a numbered list of available games:
    1. Guess the number
    2. Coin Flip
    3. Even/Odd
    4. Find the Thimble
    5. Find the Red Thread
    6. Quit
3. **User Selection:** The user enters the number corresponding to their choice.
4. **Execute Game:** The computer initiates the selected game's specific use case (e.g., "Find the Thimble").
5. **Return to Menu:** Once the specific game is finished and its individual scoreboard is shown, the computer returns to **Step 2**.
6. **Initiate Exit:** The user selects "6" to Quit.
7. **Display Final Results:** The computer displays the total number of wins and losses for both the User and the Computer across all games played.
8. **Declare Overall Winner:** The computer displays a message naming the player with the most total wins as the "Winner of the Game of Games".

4. Errors

- Invalid Selection: * If the user enters a character other than 1-6, the computer displays an error message: "Invalid choice. Please enter a number between 1 and 6.".
    - The system returns to **Step 2**.
- Overall Tie: * If both players have the same number of total wins, the computer displays: "The Game of Games ends in a draw!"

OVERALL GAME OF GAMES TESTING

SUNNY DAY PATHS

BEFORE TESTING

RUN	NUM	TEST	EXPECTED OUTCOME	OUTCOME	CORRECT	INCORRECT
	1	Main Menu, user selects a valid game to play. **Input: 1**	System initiates the "Guess the Number" game use case.			
	2	Main Menu, user selects a valid game to play. **Input: 4**	System initiates the "Find the Thimble" game use case.			
	3	Main Menu, user chooses to exit the application. **Input: 6**	System exits the menu and proceeds to Display Final Tally.			

	4	Final Tally, display total wins/losses after playing. **Input: 6**	Computer displays the total number of wins and losses for both User and Computer.			
	5	Final Winner, declare user as overall winner based on win count. **Input: (Achieve more wins than Computer, then 6)**	System displays a message naming the User as the "Winner of the Game of Games".			
	6	Final Winner, declare computer as overall winner based on win count. **Input: (Achieve fewer wins than**	System displays a message naming the Computer as the "Winner of the Game of Games".			

		Computer, then 6)				

AFTER TESTING

RUN	NUM	TEST	EXPECTED OUTCOME	OUTCOME	CORRECT	INCORRECT
	1	Main Menu, user selects a valid game to play. **Input: 1**	System initiates the "Guess the Number" game use case.	System initiates the "Guess the Number" game use case.	X	
	2	Main Menu, user selects a valid game to play. **Input: 4**	System initiates the "Find the Thimble" game use case.	System initiates the "Find the Thimble" game use case.	X	

	3	Main Menu, user chooses to exit the application. **Input: 6**	System exits the menu and proceeds to Display Final Tally.	System exits the menu and proceeds to Display Final Tally.	X	
	4	Final Tally, display total wins/losses after playing. **Input: 6**	Computer displays the total number of wins and losses for both User and Computer.	Computer displays the total number of wins and losses for both User and Computer.	X	
	5	Final Winner, declare user as overall winner based on win count. **Input: (Achieve more wins than Computer, then 6)**	System displays a message naming the User as the "Winner of the Game of Games".	System displays a message naming the User as the "Winner of the Game of Games".	X	

	6	Final Winner, declare computer as overall winner based on win count. **Input: (Achieve fewer wins than Computer, then 6)**	System displays a message naming the Computer as the "Winner of the Game of Games".	System displays a message naming the Computer as the "Winner of the Game of Games".	X	

RAINY DAY PATHS

BEFORE TESTING

RUN	NUM	TEST	EXPECTED OUTCOME	OUTCOME	CORRECT	INCORRECT
	1	Main Menu, user enters an invalid numeric choice. **Input: 9**	Error message "Invalid choice. Please enter a number between 1 and 6."			

	2	Main Menu, user enters an invalid character choice. **Input: W**	Error message "Invalid choice. Please enter a number between 1 and 6."			
	3	Final Winner, declare a draw when win counts are equal. **Input: (Achieve equal wins to Computer, then 6)**	System displays: "The Game of Games ends in a draw!".			

<u>AFTER TESTING</u>

RUN	NUM	TEST	EXPECTED OUTCOME	OUTCOME	CORRECT	INCORRECT

	1	Main Menu, user enters an invalid numeric choice. **Input: 9**	Error message "Invalid choice. Please enter a number between 1 & 6"	Error message "Invalid choice. Please enter a number between 1 & 6"	X	
	2	Main Menu, user enters an invalid character choice. **Input: W**	Error message "Invalid choice. Please enter a number between 1 and 6."	Error message "Invalid choice. Please enter a number between 1 and 6."	X	
	3	Final Winner, declare a draw when win counts are equal. **Input: (Achieve equal wins to Computer, then 6)**	System displays: "The Game of Games ends in a draw!".	System displays: "The Game of Games ends in a draw!".	X	

DROP AS BOTH A WORD AND A PDF

Meeting Notes, Project:
A Game of Games Pt. 1

Meeting Number:	Meeting Date:	Start Time:	End Time:
1	April 6, 2025	5:30 p.m.	6:00 p.m.

Attendee Name (Indicate note taker with *)	Timeliness (on time, late, left early, missing)
Aleksandra*	On time
Izzy	On time
Shivang	On time
Elias	On time

Purpose: Review individual activity diagrams and use cases for all five games in the Game of Games assignment that were done before the meeting. We also wanted to ensure consistency in format and correctness of diagrams and use cases.

Agenda:

1. Review each group member's individually prepared activity diagrams
2. Agree on formatting across all diagrams
3. Discuss and verify correctness of each game's diagram and use case
4. Resolve any discrepancies or missing elements
5. Finalize document structure for submission

Major Accomplishments: (briefly describe what was achieved the meeting)

1. We reviewed all individually prepared activity diagrams for all five games (Find the Thimble, Coin Flip, Guess the Number, Even and Odd, Find the Red Thread) and the overall Game of Games outer loop
2. Standardized the visual formatting of all diagrams to use consistent notation, label style, and layout
3. Verified correctness of game logic and completeness of diagrams, including error handling, looping, and end conditions
4. Reviewed all use cases and confirmed they aligned with the activity diagrams

Next Steps: (identify the steps to be taken next, and identify who is responsible to carrying out those steps if appropriate)

Finalize diagrams in drawing tool so they all match in format.

Meeting Notes:

Meeting began at 5:30 p.m. with all group members present. Everyone had worked individually on their activity diagrams and use cases before the meeting. This is how we divided the use cases:

Aleksandra - Activity Diagram and Use Case for Find the Thimble and Coin Flip

Elias - Activity Diagram and Use Case for Guess the Number and Even Odd

Izzy - Activity Diagram and Use Case for Find the Red Thread

Shivang - Activity Diagram and Use Case for overall Game of Games

We went through each person's diagram and talked about what needed to be changed or improved. We agreed that everyone still needs to adjust their diagrams and make sure they all look the same, using rounded rectangles for actions, diamonds for decisions, and keeping the labels consistent. We decided to use draw.io for building the diagrams. All diagrams should show gameplay between a player and a computer at a high level, without getting into implementation details.

We also looked over the use cases alongside the diagrams and made sure they match the text-based, user-vs-computer format the assignment asks for.

Meeting adjourned at 6:00 p.m.

Documents: (included or links to them)

https://docs.google.com/document/d/1SpmwvQ16cK5bPHi0YpkLdFjYhfEkNnJiSKW05phqjKY/edit?tab=t.0#heading=h.hqq2yfzb00si

Red Thread Use Cases.docx

Red Thread Activity Diagram

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DROP AS BOTH A WORD AND A PDF

Meeting Notes, Project: Game of Games Testing

Meeting Number:	**Meeting Date:**	**Start Time:**	**End Time:**
2	4/13/26	4:30	5:30

Attendee Name (Indicate note taker with *)	**Timeliness (on time, late, left early, missing)**
Elias Vasquez	On time
Izzy Pringle	On time
Shivang Mathur	On time
Aleksandra Postolov	On time

Purpose: (briefly describe general purpose of meeting expected outcomes)

Agenda: (briefly outline plan for the meeting, i.e. what actions will be taken, expected outcomes if appropriate)

- Create a sunny and rainy day tests
- Test our tests
- Update activity diagrams and use cases

Major Accomplishments: (briefly describe what was achieved the meeting)

- Completed the agenda
- Worked together and communicated clearly
- Created successful tests

Next Steps: (identify the steps to be taken next, and identify who is responsible to carrying out those steps if appropriate)

- Start to code?
- To be determined

Meeting Notes: (Notes are not a direct transcription, but a condensed, as I would say somewhat briefly but completely, record of what was said and happened at the meeting. A major purpose of project notes is to have an accurate record of information, discussion points and decision influences and decisions made that can be referenced later if needed)

We met on Monday, 4/13, at 4:30 to work on Game of Games Testing and everyone was on time. We started by going through all the directions and understanding what was being asked of us. Next, we decided to create our sunny day and rainy day test cases for each of the games that we created. Then, we went through our Use Case and Activity Diagram document to test our test cases and see if we needed to make any updates to our original work. From there, we made changes and updated our document. Finally, we created one final document where we pasted in all our work: Use Cases, Activity Diagrams, and Testing. We left at 5:30 and agreed that we would finish anything that we didn't complete together on our own and submit once everyone was done.

Documents: (included or links to them)

GameOfGamesTestingEliasIzzyShivangAleksandra - Google Docs